

Denoising Covariance Matrices, Portfolio Instability & ETF Construction

Advanced Lecture Notes on Financial Machine Learning & ETF Mechanics

AI Finance Seminar

Sogang University

September 2026

Table of Contents

- 1 ETF Construction & Synthetic Replication
- 2 Denoising & Detoning Covariance Matrices
- 3 Portfolio Weight Instability & Markowitz's Curse
- 4 Summary & Key Takeaways

Section 1: ETF Construction & Replication Mechanics

Core Objective

Understand how modern Exchange-Traded Funds (ETFs) are engineered, how physical and synthetic replication models operate, and how primary market arbitrage preserves liquidity.

Key topics in this section:

- The 5-Layer ETF Construction Stack
- Physical (Full/Sampled) vs. Synthetic (Swap-Based) Replication
- Primary Market Mechanics: Rule 6c-11, Custom Baskets & Tax Efficiency
- Tracking Quality Metrics: Tracking Difference vs. Tracking Error

The 5-Layer ETF Construction Stack

① Layer 1: Index Methodology

- Eligibility rules, free-float adjustments, and weighting schemes (market-cap, equal, factor).
- Regulatory capping rules (e.g., UCITS 5/10/40 or US RIC 25/5/50) to limit concentration.

② Layer 2: Portfolio Replication Strategy

- Full Replication, Stratified Sampling/Optimization, or Synthetic Total Return Swaps.

③ Layer 3: Wrapper & Legal Regime

- Regulatory structures (US 1940 Act, UCITS) enforcing leverage and counterparty limits.

④ Layer 4: Primary Market & Liquidity Mechanics

- Authorized Participants (APs) executing creation/redemption in creation units.

⑤ Layer 5: Risk & Implementation Controls

- Managing reconstitution drag, bid-ask spread slippage, and securities lending splits.

Physical vs. Synthetic (Swap-Based) Replication

Physical Replication

- Holds actual underlying index securities.
- **Full Replication:** Holds all constituents; ideal for liquid mega-caps.
- **Stratified Sampling:** Holds a representative factor-matched subset; necessary for illiquid bonds and emerging markets.
- Incurs trading drag, illiquidity costs, and sampling errors.

Synthetic Replication

- Holds a substitute/collateral basket and enters a **Total Return Swap (TRS)** with an investment bank.
- Bank guarantees exact index performance in exchange for substitute basket yield.
- Eliminates tracking error on hard-to-access markets.
- Tax advantages: e.g., US Section 871(m) withholding tax exemption on US equities.

Primary Market Arbitrage & Tracking Quality

Rule 6c-11 & Custom Baskets

SEC Rule 6c-11 allows US ETFs to use **non-pro-rata custom creation/redemption baskets**.

- Enables APs to exchange specific illiquid or high-basis securities in-kind.
- Improves tax efficiency under IRC §852(b)(6) by flushing out low-basis shares without triggering capital gains taxes.

Tracking Difference (TD):

$$TD = R_{\text{ETF}} - R_{\text{Index}}$$

Measures cumulative return gap over time (driven by Expense Ratio, tax drag, lending yield).

Tracking Error (TE):

$$TE = \sqrt{252} \cdot \sigma(R_{\text{ETF},t} - R_{\text{Index},t})$$

Annualized standard deviation of daily return differences (measures volatility of tracking mismatch).

Section 2: Denoising Covariance Matrices (book2, Ch. 2)

The Fundamental Challenge

Empirical covariance matrices Σ calculated from finite time series are dominated by estimation noise. Inverting noisy matrices severely destabilizes portfolio optimization.

In this section, we cover:

- Random Matrix Theory (RMT) and the Marcenko–Pastur Theorem
- Separating Noise Eigenvalues from True Economic Signals
- The Constant Residual Eigenvalue Denoising Method
- Detoning: Removing the Dominant Market Component

Random Matrix Theory: The Marcenko–Pastur Theorem

For a $T \times N$ random matrix of IID observations with variance σ^2 , as $T, N \rightarrow \infty$ with $q = \frac{T}{N} \geq 1$, the theoretical PDF of eigenvalues λ of its correlation matrix is:

$$f(\lambda) = \frac{q}{2\pi\sigma^2\lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} \quad \text{for } \lambda \in [\lambda_-, \lambda_+]$$

where the theoretical upper and lower noise bounds are:

$$\lambda_{\pm} = \sigma^2 \left(1 \pm \sqrt{\frac{1}{q}} \right)^2$$

Core Insight

Eigenvalues falling within $[\lambda_-, \lambda_+]$ represent **pure random noise**. Only eigenvalues exceeding λ_+ ($\lambda > \lambda_+$) represent genuine **economic signals**.

Denoising Method: Constant Residual Eigenvalues

Unlike Ledoit–Wolf shrinkage (which shrinks all elements blindly), RMT denoising preserves signal eigenvectors while scrubbing noise:

- 1 **Eigendecomposition:** Diagonalize the empirical correlation matrix $\mathbf{C} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^T$.
- 2 **Identify Noise Cutoff:** Fit the Marcenko–Pastur PDF via Kernel Density Estimation (KDE) to find λ_+ . Let i be the number of signal eigenvalues ($\lambda_k > \lambda_+$).
- 3 **Replace Noise Eigenvalues:** Replace all $N - i$ noise eigenvalues ($\lambda_k \leq \lambda_+$) with their average:

$$\bar{\lambda}_k = \frac{1}{N - i} \sum_{k=i+1}^N \lambda_k \quad \text{for } k = i + 1, \dots, N$$

- 4 **Reconstruct Matrix:** Form $\mathbf{C}_{\text{denoised}} = \mathbf{W}\tilde{\mathbf{\Lambda}}\mathbf{W}^T$ and rescale main diagonal to 1.

Detoning: Removing the Market Factor

The Market Component Problem

The largest eigenvalue $\lambda_{\max} (\lambda_1)$ in financial correlation matrices represents a market-wide factor where all assets move together. High baseline market correlation masks micro-structures and sector clusters.

The Detoning Procedure:

- Subtract the first principal component (associated with λ_1 and eigenvector $\mathbf{W}_1$) from the denoised correlation matrix:

$$\mathbf{C}_{\text{detoned}} = \mathbf{C}_{\text{denoised}} - \lambda_1 \mathbf{W}_1 \mathbf{W}_1^T$$

- Rescale diagonal back to 1s.
- **Result:** Unsupervised clustering algorithms can now detect subtle, sector-specific alpha relationships without being blinded by market-wide movements.

Section 3: Portfolio Weight Instability & Markowitz's Curse

The Core Dilemma

Why do traditional mean-variance optimizers fail so spectacularly in real life? Because small estimation errors in non-invertible or ill-conditioned covariance matrices cause catastrophic weight explosion.

Topics in this section:

- The Mathematics of Matrix Inversion Explosion
- Markowitz's Curse & Condition Number Analysis
- Solutions: Hierarchical Risk Parity (HRP) & Nested Clustered Optimization (NCO)

Markowitz's Curse & Condition Number Explosion

Optimal portfolio allocation requires solving $w^* = \frac{\mathbf{V}^{-1}\mathbf{a}}{\mathbf{a}'\mathbf{V}^{-1}\mathbf{a}}$.

The numerical sensitivity of the inverse covariance matrix $\mathbf{V}^{-1}$ is measured by its **Condition Number**:

$$\text{Condition Number} = \frac{\lambda_{\max}}{\lambda_{\min}}$$

Markowitz's Curse

- As asset correlation $\rho \rightarrow 1$, the matrix determinant $|\mathbf{V}| \rightarrow 0$ and $\lambda_{\min} \rightarrow 0$.
- The Condition Number approaches infinity, causing $\mathbf{V}^{-1}$ entries to **explode**.
- **The Paradox:** The more correlated assets are, the greater the economic need for diversification—yet the more correlated assets are, the more unstable Markowitz weights become!

Solving Instability: HRP & NCO

López de Prado presents two machine learning solutions to eliminate portfolio weight instability:

Hierarchical Risk Parity (HRP)

- Uses graph-theoretic tree clustering on correlation distance.
- Allocates weights top-down using recursive bisection.
- **No Matrix Inversion Required!**
- Works even if $\mathbf{V}$ is singular or non-positive definite.

Nested Clustered Optimization (NCO)

- Clusters assets into K sub-groups using Silhouette scores.
- Performs intra-cluster optimization within each cluster separately.
- Performs inter-cluster optimization across cluster centroids.
- **Isolates Condition Numbers** inside clusters, stopping instability propagation.

Summary: The Complete Engineering Framework

- 1 **ETF Mechanics:** Use physical sampling or synthetic swaps to replicate indices, and leverage custom creation baskets (§852(b)(6)) for maximum tax efficiency.
- 2 **Denoising Covariance:** Apply Random Matrix Theory (Marcenko–Pastur) to replace noise eigenvalues while preserving true signal.
- 3 **Detoning Factor:** Strip out $\lambda_{\max}$ to uncover micro-structure sector clusters.
- 4 **Conquering Weight Instability:** Stop relying on raw Markowitz inversions. Use HRP or NCO to eliminate condition number explosions and achieve robust out-of-sample portfolio stability.

Q&A

Thank You! Questions & Discussion